

Problem Set - Section 5.2

1) $A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$

~~$\det(A) = \dots + [(1)(1)(1)]$~~

Big Formula Method

	R_1	R_2	R_3	
$(C_1, C_2, C_3) (1, 2, 3)$				$\rightarrow (+)(1)(1)(1) = +1$
$(2, 3, 1)$				$\rightarrow (+)(2)(2)(3) = +12$
$(3, 1, 2)$				$\rightarrow (+)(3)(3)(2) = +18$
$(1, 3, 2)$				$\rightarrow (-)(1)(2)(2) = -4$
$(2, 1, 3)$				$\rightarrow (-)(2)(3)(1) = -6$
$(3, 2, 1)$				$\rightarrow (-)(3)(1)(3) = -9$

$$\begin{aligned} \det(A) &= +1 + 12 + 18 - 4 - 6 - 9 \\ &= +31 - 19 \\ &= \underline{+12} \end{aligned}$$

$B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 6 & 7 \\ 5 & 6 & 7 \end{bmatrix}$

~~$\det(B) =$~~

$(1, 2, 3) \rightarrow$	$(+)(1)(4)(7) = +28$
$(1, 3, 2) \rightarrow$	$= -24$
$(2, 1, 3) \rightarrow$	$= -56$
$(2, 3, 1) \rightarrow$	$= +40$
$(3, 1, 2) \rightarrow$	$= +72$
$(3, 2, 1) \rightarrow$	$= -60$

$\det(B) = \underline{+64}$

Row 1 + Row 2
= Row 3

∴ Linearly Dep.
∴ $\det(B) = 0$

$\det(C) = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$

$$\begin{aligned} &= (+)(0) + (-)(0) + (-)(0) + (+)(0) + \\ &\quad (+)(0) + (-)(1) \end{aligned}$$

$$= 0 + 0 + 0 + 0 + 0 - 1$$

$= \underline{-1}$

so C has independent rows

4) Non-zero choices:

$$(4, 3, 2, 1) = (+)(1) = +1$$

$$(1, 3, 2, 4) = (-)(1) = -1$$

$$\det(A) = (+1) + (-1) = 0$$

$$\det(B) = (+)(64) + (-)(16) = 48$$

7) $\det P = +1$

positive permutations

$$\text{Permutations} = n! = 5! = 120$$

$$\text{Positive Half} = \frac{120}{2} = 60$$

$$(1, 2, 3, 4, 5)$$

$$(2, 1, 4, 3, 5)$$

$$(2, 3, 4, 1, 5)$$

$$(5, 3, 4, 1, 2)$$

$$(5, 4, 3, 2, 1)$$

$$(3, 5, 4, 1, 2) \times$$

$$(5, 1, 2, 3, 4)$$

10)

$$11) C_{ij} = (-1)^{i+j} \cdot \det M_{ij}$$

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$C = \begin{bmatrix} d & -c \\ -b & a \end{bmatrix}$$

$$C_{11} = (-1)^2 (d)$$

$$C_{12} = (-1)^3 (c)$$

$$C_{21} = (-1)^3 (b)$$

$$C_{22} = (-1)^4 (a)$$

$$AC = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} d & -c \\ -b & a \end{bmatrix} = \begin{bmatrix} ad-b^2 & ab-ac \\ cd-bd & ad-c^2 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 0 & 0 \end{bmatrix}$$

$$\det B = b_{11}C_{11} + b_{12}C_{12} + b_{13}C_{13}$$

$$= (1)((-1)^2(0)) +$$

$$(-2)((-1)^3(-(6)(7))) +$$

$$(3)((-1)^4(-(5)(7)))$$

$$= (0) + (+84) + (-168)$$

$$= 0 + 84 - 168$$

$$= 84 - 168$$

$$= -84$$

{ Forget to make cofactor matrix D

↳ D allows to read off C_{ij}

14) $(2, 1, 4, 3)$

Select column 2, then col 1, then col 4, then col 3.

$$\left. \begin{array}{l} C_3 \rightarrow \det = 0 \\ C_1 \rightarrow 0 \\ C_2 \rightarrow 1 \\ C_4 \rightarrow 1 \end{array} \right\} \begin{array}{l} \text{Requires } n \text{ to be even.} \\ \text{If } n \text{ is odd } \det A = 0. \end{array}$$

Permutations: $C_{ij} = (-1)^{i+j} M_{ij}$

Reason

Row exchanges is $\frac{n}{2}$
 $(4 \rightarrow 2 \text{ row exchanges})$

so $C_n = (-1)^{n/2}$

17) $|B_4| = 2|B_3| - |B_2| = 1$

$$\begin{aligned} |B_4| &= b_{41}C_{41} + b_{42}C_{42} + b_{43}C_{43} + b_{44}C_{44} \\ &= (0)C_{41} + (0)C_{42} + (-1)C_{43} + 2C_{44} \\ &= 2C_{44} - C_{43} \end{aligned}$$

$$= 2[(-1)^8(B_3)] - [(-1)^7(B_2)]$$

$$= 2|B_3| + \det \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$$

$$= 2|B_3| + \det \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$$

$$= 2|B_3| + (-1)^6 |B_2|$$

$$= 2|B_3| + a_{33}C_{33}$$

$$= 2|B_3| + (-1)(-1)^6(|B_2|)$$

$$= 2|B_3| - |B_2|$$

20) $G_2 = \begin{vmatrix} 0 & 1 \\ 0 & 1 \end{vmatrix} = (0)(0) - (1)(1) = -1$

$$C_{13} = \begin{vmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{vmatrix} = a_{13}C_{13} + a_{23}C_{23}$$

$$= (1)[(-1)^4 \begin{vmatrix} 1 & 0 \\ 1 & 1 \end{vmatrix}] + (1)[(-1)^5 \begin{vmatrix} 0 & 1 \\ 1 & 1 \end{vmatrix}]$$

$$= (1) + (-1)(-1)$$

$$= 2$$

$G_4 = \begin{vmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{vmatrix} \Rightarrow \begin{vmatrix} 0 & -1 & -1 & -1 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{vmatrix}$

$\Rightarrow \begin{vmatrix} -1 & -1 & -1 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{vmatrix}$

$\Rightarrow \begin{vmatrix} -1 & -1 & -1 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{vmatrix}$

$\Rightarrow \begin{vmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{vmatrix}$

$\Rightarrow \begin{vmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{vmatrix}$

$\Rightarrow \begin{vmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{vmatrix}$

Exchanges
$\Rightarrow$ 3

$\Rightarrow \begin{vmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{vmatrix}$

$\Rightarrow \begin{vmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{vmatrix}$

$\det = 0$

$$G_u = \begin{vmatrix} 0 & 1 & -1 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{vmatrix} \Rightarrow \begin{vmatrix} 0 & 1 & -1 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{vmatrix} \Rightarrow \begin{vmatrix} 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{vmatrix}$$

$$\Rightarrow \begin{vmatrix} 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{vmatrix} \Rightarrow \begin{vmatrix} 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{vmatrix}$$

$$\Rightarrow \begin{vmatrix} 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{vmatrix} \Rightarrow \begin{vmatrix} 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{vmatrix}$$

RE: 1

20 con.) $G_4 = \begin{vmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{vmatrix}$ RE # 1

$$\Rightarrow \begin{vmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \end{vmatrix} \Rightarrow \begin{vmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{vmatrix} \Rightarrow \begin{vmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix}$$

$$\Rightarrow \begin{vmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix} \Rightarrow \begin{vmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 2 \end{vmatrix} \Rightarrow \begin{vmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 3 \end{vmatrix}$$

~~det~~ $G_4 = (-1)^1 [1(-1)(-1)(3)] = -3$ ✓

$$G_2 = -1$$

$$G_3 = 2$$

$$G_4 = -3$$

~~det~~ G_n

$$G_n = (-1)^{(n-1)}(n-1)$$
 ✓

23) a) ~~First statement only involves the O.C.~~
 First statement $\begin{vmatrix} A & B \\ 0 & D \end{vmatrix} = \det(A)\det(D)$ as it is

b) $\begin{vmatrix} A & B \\ C & 0 \end{vmatrix} \Rightarrow \begin{vmatrix} A \end{vmatrix}$

upper triangular
 → any results used in B together with zero is zero and has no effect

∴ Block matrices behave as scalars for multiplication & addition but not for determinants

26) $A_{(m \times n)}, B_{(m \times n)} \Rightarrow \det(M) = \det(AB)$

$$M = \begin{bmatrix} 0 & A \\ -B & I \end{bmatrix} = \begin{bmatrix} AB & A \\ 0 & I \end{bmatrix} \begin{bmatrix} I & 0 \\ -B & I \end{bmatrix}$$

~~$A = \begin{pmatrix} 1 & 2 & 3 \end{pmatrix}$
 $B = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$~~ $\det M = |AB|$

$$|M| = \begin{vmatrix} 0 & A \\ -B & I \end{vmatrix}$$

$A = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$
 $B = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$

$$M = \begin{bmatrix} 0 & 1 & 2 & 3 \\ -1 & 0 & 0 & 0 \\ -2 & 0 & 1 & 0 \\ -3 & 0 & 0 & 1 \end{bmatrix}$$

~~$\det AB$~~
 $= \det \begin{pmatrix} 14 \end{pmatrix}$
 $= 14$ ✓

$$|M| = \begin{vmatrix} 0 & 1 & 2 & 3 \\ -1 & 0 & 0 & 0 \\ -2 & 0 & 1 & 0 \\ -3 & 0 & 0 & 1 \end{vmatrix}$$

$$\Rightarrow \begin{vmatrix} 0 & 1 & 2 & 3 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -3 & 0 & 0 & 1 \end{vmatrix} \begin{matrix} (-3) - 3(-1) \\ (0) - 3(-2) \\ (1) - 3(-3) \end{matrix}$$

$$\Rightarrow \begin{vmatrix} 0 & 1 & 2 & 3 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 6 & 10 \end{vmatrix}$$

$$\Rightarrow \begin{vmatrix} 0 & 1 & 2 & 3 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 10 \end{vmatrix} \begin{matrix} (6) - (\frac{6}{5})(3) \\ (10) - (\frac{6}{5})(6) \\ 10 - \frac{36}{5} \end{matrix}$$

$$\det M = (-1) \left[(-1)(1)(5) \left(\frac{14}{5} \right) \right] = (-1) \left[-14 \right] = 14$$

$$\det M = a_{31}(c_{31} + a_{32}(c_{32} + a_{33}(c_{33} + a_{34}(c_{34})))$$

$$= (-3)(c_{31} + (1)(c_{34})) = (-3)(-1)^4 \left(\begin{vmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \end{vmatrix} \right) + (1)(-1)^7 \left(\begin{vmatrix} 0 & 2 \\ -3 & 0 \end{vmatrix} \right)$$

$$= (-3) \begin{vmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \end{vmatrix} - \begin{vmatrix} -1 & 1 & 0 \\ 0 & -3 & 0 \end{vmatrix}$$

$$= (-3) \left[-2 \right] - \begin{vmatrix} -1 & 1 & 0 \\ 0 & 0 & 6 \end{vmatrix}$$

$$= (6) - (-1) \begin{vmatrix} -1 & 1 & 0 \\ 0 & 0 & 6 \end{vmatrix}$$

$$= (6) - (-1)(-6)$$

$$= (6) - (6) = 0$$

∴ $\det M = \det AB$ ✓

$$29) E_4 = \begin{vmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{vmatrix}$$

Big formula $n! = (4)! = 24$ terms
 $\hookrightarrow$ 5x terms are non-zero.

$$E_4 = \begin{vmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{vmatrix}$$

$$(1, 2, 3, 4) \rightarrow (-1)^0 [1 \cdot 1 \cdot 1 \cdot 1] = +1$$

$$(2, 1, 3, 4) \rightarrow (-1)^{+1} [1] = -1$$

$$(1, 3, 2, 4) \rightarrow (-1)^{+1} [1] = -1$$

~~(2, 3, 1, 4)~~

~~(1, 4, 2, 3)~~

$$(1, 2, 4, 3) \rightarrow (-1)^{+1} [1] = -1$$

$$(2, 1, 4, 3) \rightarrow (-1)^2 [1] = +1$$

$$\therefore E_4 = (1) + (-1) + (-1) + (-1) + (1) \\ = \underline{-1} \quad \checkmark$$